ENGINEERING SPECIFICATION

Rule Evaluation Engine

Consolidated design. Parameter classification, score and weight composition, override precedence, and the evaluation pipeline.

Version 0.1 · Ruleset 1.0.0 · Procedure TKR

Consolidates the decisions in the PRD, HLD and LLD

17 September 2026

SECTION 1

Overview

The engine takes one patient submission, compares it against thresholds tuned to that patient's own discharge baseline and post-operative day, and returns a single tier plus the ranked reasons behind it. It is the one component in the system capable of causing harm, and the one that must be auditable by a clinician.

Two stages, in strict order

Hard overrides run first and bypass scoring entirely. A weighted sum runs second, across everything that did not override.

A single weighted total on its own is insufficient for three reasons. A genuinely critical individual finding can be diluted below threshold by good scores elsewhere. A numeric total gives a nurse nothing actionable. And when false positives appear, they cannot be traced back to a specific weight. The two-stage shape mirrors established early-warning score practice, where any single parameter at its maximum escalates irrespective of the aggregate.

Contract

```
evaluate(reading, window, config) -> Result
```

Property	Requirement
Pure	No input or output of any kind. Reading, window, baseline and config in; tier and reasons out. This is what makes it exhaustively testable without a database, a device or a patient.
Deterministic	Identical inputs and ruleset version must reproduce the identical tier and reasons. A stored decision is reconstructable indefinitely.
Synchronous	Runs in the request path. The device must learn immediately whether to display emergency guidance.
Dual-invoked	Called from the check-in endpoint on submission, and from the alert worker with a synthetic all-unknown reading when a check-in is missed. One implementation, two call sites. A parallel version in the worker is a defect, because the two diverge at the first threshold change.
Config-driven	Every threshold, band, weight and detector lives in versioned configuration. The version is stamped on every evaluation.

Output tiers

Tier	Reached by	Consequence
emergency	Override only	Patient-side guidance shown immediately. Care team notified in parallel. Never queued.
critical	Override or total	Alert created. Push and SMS to on-call nurse. Escalation timer starts.
review	Total only	Enters the dashboard queue. No interrupting notification.
normal	Default	Logged. No action.
missed	Absence sweep	Distinct status. Raised by the worker, never by ingest.

SECTION 2

Parameter classification by value type

The design rests on one normalisation rule: **every parameter, whatever its native value type, resolves to an integer 0 to 3 before anything downstream touches it.** That is what allows a yes/no answer and a 0-to-10 slider to be compared, weighted and summed at all.

2.1 The five input classes

Class	Config key	Parameters	How the 0–3 score is derived
`CONTINUOUS FIXED`	`bands`	temp_c, spo2	The raw value is located in a static band list. Band boundaries never change with time or patient. Used where the clinically dangerous value is the same on day 1 and day 40.
`CONTINUOUS DAY-INDEXED`	`bands_by_day`	pain	A separate band list per phase of recovery, keyed by inclusive day range. The same reported value scores differently by day, encoding the expected recovery corridor without any history lookup. The cheapest defence against week-one false positives.
`CONTINUOUS RELATIVE`	`relative_to_baseline`	hr	Bands are read against the *delta* from the patient's discharge baseline, not the absolute value. A pulse of 96 means nothing alone; 96 against a baseline of 70 is a delta of 26.
`BINARY`	`binary_score`	calf_pain, breathless, drainage, redness, chills, weight_bear	A declared constant. A true answer scores exactly that value; false scores 0. Declaring `binary_score: 3` makes the parameter an override by construction, with no separate override list to drift out of sync.
`ORDINAL`	`milestone`	knee_bend, walk_aid	Options carry an `ordinal_rank` in the parameter registry, so walker < crutches < cane < none is genuinely comparable. The reported rank is compared against the rank expected by that day; falling short scores the declared value.

> **SIXTH CLASS — DERIVED, NOT AN INPUT**
>
> Trend signals are computed from the window rather than read off the submission, but they are **first-class signals flowing through the identical band-and-weight pipeline**, not an additive bonus bolted on at the end. This matters twice over: a trend can trigger an override, and a trend appears in the reasons list the nurse actually reads. Section 6 specifies the five detectors.

2.2 Band mechanics

Every band is a triple `[lower, upper, score]`, **half-open** — lower inclusive, upper exclusive.

```
bands: [[0,37.5,0], [37.5,38.0,1], [38.0,38.5,2], [38.5,99,3]]
        37.4 -> 0     37.8 -> 1       38.2 -> 2      38.6 -> 3
```

The final band must carry a **sentinel upper bound** — 99 for temperature, 999 for a heart-rate delta. A top band written as `[9,10,3]` on a 0-to-10 pain scale cannot match the value 10, and because the engine emits *no signal at all* for an unassessable value rather than a zero, the worst reading a patient can give would produce silence. That is the most dangerous failure the config can contain, and it arrives through an off-by-one.

> **TWO CONVENTIONS COEXIST — DOCUMENT THEM**
>
> Day ranges in `bands_by_day` are **inclusive at both ends** (`"4-10"` covers days 4 through 10). Band triples are **inclusive-exclusive**. Both appear within the same rule, and mixing them up is a silent miscalibration rather than an error.

2.3 Name versus parameter

A rule carries both a `name` (the signal identifier that appears in reasons, dedup keys and snapshot tests) and a `parameter` (the field it reads). They are frequently identical, which hides why both exist. They diverge where it counts: `name: heart_rate` reads `parameter: hr`, and critically `name: pain_rising` also reads `parameter: pain`. One parameter, several signals — which is precisely why the per-parameter cap in Section 3 keys on `parameter` and not on `name`.

SECTION 3

Composite weight and threshold

3.1 Score and weight are different things

Quantity	Range	Answers
Band score	0 – 3	How bad is this parameter right now?
Weight	0 – 10	How much does this parameter matter relative to the others?
Points	score × weight	What this signal contributes to the composite total.

Temperature carries weight 3 and pain weight 2, so a mildly raised temperature contributes more than mildly raised pain even though both score 1. Pain is weighted moderately on purpose: it is subjective and self-reported.

3.2 The per-parameter cap

```
max_points_per_parameter: 6
```

A high-and-rising pain reading scores twice — once on its band, once on its trend. That is usually correct but compounds fast. The cap keeps the highest-scoring signals for a parameter and discards the remainder, so no single vital can dominate the composite on its own. It applies **after** all signals are scored and **before** the override checks.

3.3 Maximum contribution per parameter

Because band 3 always triggers an override, the greatest a parameter can contribute *to the weighted sum* is band 2 multiplied by its weight. Parameters declaring `binary_score: 3` therefore never appear in a sum at all — they either override or contribute nothing.

Signal	Class	Weight	Max summed band	Max points	Effect
temperature	Continuous fixed	3	2	**6**	Half of critical on its own
redness	Binary	3	2	**6**	Half of critical on its own
pain (band + trend)	Day-indexed	2	2 + 2	**6**	Capped down from 8
heart_rate	Continuous relative	2	2	**4**	Clears review alone
spo2	Continuous fixed	2	2	**4**	Clears review alone
wound_drainage	Binary	2	2	**4**	Clears review alone
slow_burn	Trend	2	2	**4**	Clears review alone
mobility_regression	Trend	1	2	**2**	Contributory only
knee_bend	Ordinal	1	2	**2**	Contributory only
walk_aid	Ordinal	1	1	**1**	Contributory only
persistent_abnormal	Trend	1	1	**1**	Contributory only
calf_pain, chills, weight_bear, breathless	Binary	3	–	**–**	Always override; never summed
fever_returned	Trend	3	–	**–**	Always override; never summed

3.4 Tier thresholds

```
tiers: { review_at: 5, critical_at: 12 }
```

Reading the calibration off the table above gives the minimum combinations that reach each tier, which is the most useful sanity check to put in front of a surgeon:

Tier	Cutoff	Smallest combinations that reach it
review	5	Temperature band 1 (3) plus pain band 1 (2). Or drainage alone (4) plus any 1-point signal. Deliberately easy to reach — review costs a nurse thirty seconds.
critical	12	Temperature band 2 (6) plus redness (6). Or pain capped (6) plus heart rate band 2 (4) plus any 2-point signal. Requires genuine multi-system involvement.

The gap between 5 and 12 is the design's main calibration lever. Narrowing it converts review-tier items into pages; widening it sends more to the next shift. Tune this before touching individual weights, because it moves the whole distribution at once and is far easier to reason about.

> **GUARDRAIL**
>
> Target five or fewer alerts per nurse per day in steady state. Above that, thresholds are miscalibrated and the queue gets ignored however correct each individual decision is.

SECTION 4

Evaluation pipeline

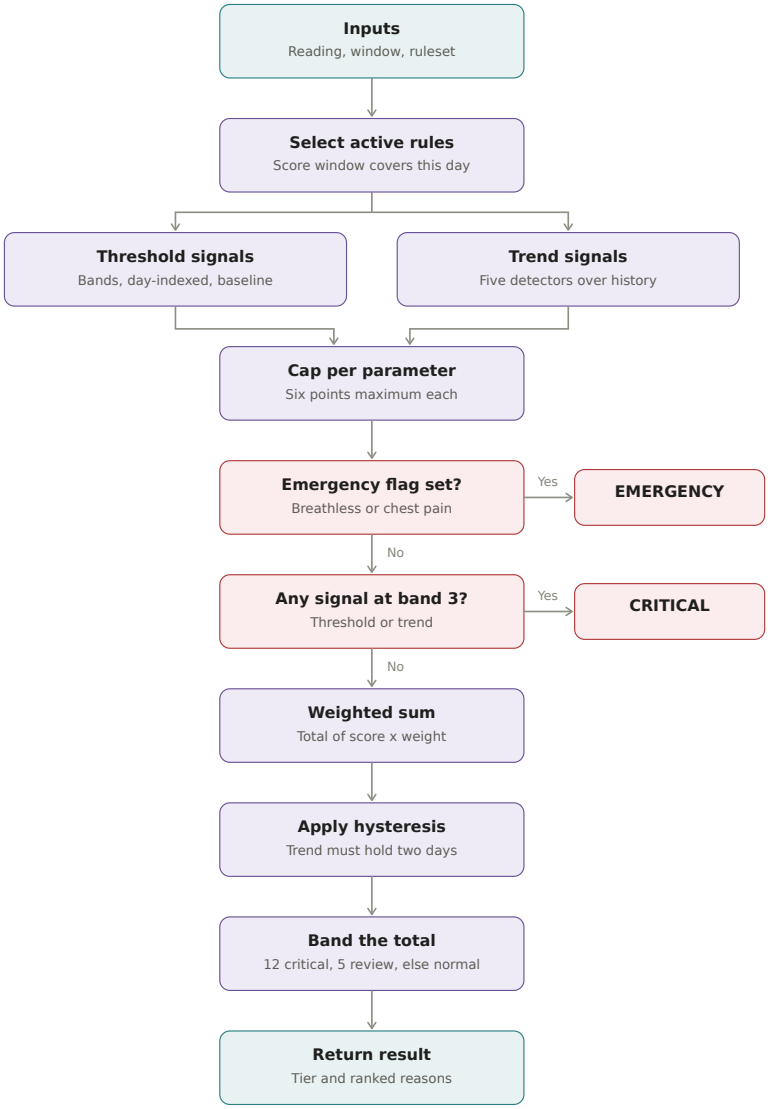


Figure 1. Evaluation pipeline. Overrides exit right and never reach the weighted sum.

Order is load-bearing

1. **Rule selection precedes scoring.** Only rules whose *score* window covers this post-operative day participate. Collected-but-not-yet-scored values are stored and ignored.
2. **Trends receive the threshold signals.** Detectors such as `persistent_abnormal` need to know what is currently abnormal, so trend scoring runs after threshold scoring, not alongside it.
3. **The cap applies before the overrides.** Capping is about the composite total, not about suppressing a maximal signal.
4. **Emergency precedes band 3.** Both are overrides, but only emergency triggers patient-facing guidance, so it must be checked first.
5. **Hysteresis applies only to the weighted path.** An override is never delayed for confirmation.

SECTION 5

Overrides and trend detectors

5.1 Override precedence

Order	Condition	Result	Rationale
1	`breathless` or `chest_pain` true	emergency	The only signal that shows guidance on the patient's own screen and never enters a queue. Also evaluated client-side so it works offline.
2	Any signal scores 3	critical	Declarative. `binary_score: 3` and trend `score: 3` are how a rule announces it is sufficient alone, with no separate override list to fall out of sync.
3	Total ≥ `critical_at`	critical	The combination case. No individual signal is maximal.
4	Total ≥ `review_at`	review	Queued, not notified.

5.2 Trend detectors

Detector	Signal	Score	Question it asks
`delta_vs_rolling_mean`	pain_rising	banded 0–3	Is today's value above this patient's own recent average by more than the band threshold? Silent until `min_observations` is met.
`reemergence`	fever_returned	3 — overrides	Did this settle for `normal_days_required` days and then come back? The highest-yield detector in the set: a temperature that settles for two days and climbs again on day six is a far stronger infection signal than the same reading on day one, and no absolute threshold catches it.
`monotonic_rise`	slow_burn	2	Has it climbed steadily across the window, totalling at least `min_total_rise`? Catches a creep that never crosses a hard threshold.
`drop_from_personal_best`	mobility_regression	2	Has it fallen below `drop_fraction` of the best this patient has achieved? Requires whole-window state, not a rolling mean.
`streak`	persistent_abnormal	1	Has anything been abnormal for `min_days` consecutively? `parameter: any`, so it spans all signals rather than one.

5.3 Windows — collect versus score

Two independent windows per rule, with a database constraint enforcing that the score window sits inside the collect window.

Window	Consumed by	Meaning
`collect_from_day` / `collect_to_day`	Form schema endpoint	What the patient is asked.
`score_from_day` / `score_to_day`	Rule evaluation	What the engine acts on.

Drainage is the motivating case: asked from day 0, scored from day 5. Drainage *duration* is the predictor rather than its presence, so hiding the question until day 5 would discard the very thing that matters and give the trend detectors a cold start on the parameter where persistence counts most.

The form gate is client-side only. Out-of-window values still arrive through nurse entry during a phone call, through timezone disagreement at a day boundary, through an offline queue crossing a ruleset change, and through direct API calls. **The engine must therefore enforce its own windows and never assume the form did.**

SECTION 6

Safety rules and worked examples

6.1 Rules that prevent false confidence

Rule	Reason
A detector below `min_observations` emits nothing	Never a zero. Zero means assessed and normal, which is a lie during warm-up and would mask real problems in the first days.
Missing data is never scored	A skipped reading is unknown, not reassuring. Coverage is tracked separately and a low-coverage evaluation is marked low-confidence.
The window is indexed by post-operative day	With an explicit null for gaps — never by "the last three submissions". Otherwise a patient who skips a week has day 9 silently compared against day 2, and every trend detector quietly lies.
Hysteresis on non-override trends	A trend signal must hold across two consecutive evaluations before it can raise the tier. Trend states persist for days and would otherwise flap across band boundaries.
Return reasons, never the number	The dashboard shows "fever returned, redness spreading, pain rising". The score is internal machinery for ordering the queue and is never exposed to any client.
Deduplicate on the reason set	`reason_key` is a stable hash of the sorted contributing signal names. An identical key within 24 hours creates no new alert unless the tier increases. Without this, a persistent fever re-alerts every morning until the queue becomes wallpaper.

6.2 Worked example — the combination case

Day 9. Pain 7 against a three-day mean of 4, temperature 37.8, pulse 96 on a baseline of 70, redness spreading, steps down from a personal best of 500 to 150.

Signal	Class	Band	Weight	Points	Note
pain	Day-indexed	2	2	4	7 falls in [6,8) on the day 4–10 corridor
pain_rising	Trend	2	2	4	delta 3 falls in [3,4)
	pain parameter subtotal 8, capped			**6**	Cap visible in practice
temperature	Continuous fixed	1	3	**3**	37.8 falls in [37.5,38.0)
heart_rate	Continuous relative	2	2	**4**	delta 26 falls in [20,30)
redness	Binary	2	3	**6**	binary_score 2
mobility_regression	Trend	2	1	**2**	70% drop exceeds the 50% fraction
			Composite total	**21**	**≥ 12, so critical**

No signal reached band 3. This escalates purely on the weighted total, which is the case a pure override model cannot detect and the reason the second stage exists at all.

6.3 Worked example — the same readings, day 2

Identical pain of 7 and temperature of 37.8, nothing else abnormal.

Signal	Class	Band	Weight	Points	Note
pain	Day-indexed	0	2	**0**	7 falls in [0,8) on the day 0–3 corridor
pain_rising	Trend	–	2	**0**	Pain is descending; no delta
temperature	Continuous fixed	1	3	**3**	Same band as day 9
			Composite total	**3**	**< 5, so normal**

The same reported numbers, seven days apart, produce critical and normal respectively. That difference comes entirely from the day-indexed corridor and the trend layer, and it is the single clearest demonstration of why flat population thresholds are unusable in the first post-operative week.

CLINICAL GOVERNANCE

Every band boundary, weight and tier cutoff in this specification is a starting point requiring surgeon approval before any patient use. Values vary by surgeon, implant and institution. Continuous integration runs thirteen synthetic fixture patients on every change to the evaluator or an active ruleset; any change that alters a fixture's first-escalation day fails the build and requires explicit clinical re-approval recorded as a new ruleset version.